Nguyen Song Thien Long

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Research Interests

My research interests lie in **Natural Language Processing (NLP)**, with a primary focus on Information Retrieval (IR), Question Answering (QA), and Large Language Models (LLMs). My work centers on Retrieval-Augmented Generation (RAG), hybrid lexical–semantic retrieval, and knowledge graph-grounded, multi-hop reasoning in both agentic and standalone LLM systems. I am especially interested in making these methods efficient and trustworthy, including in low-resource and multilingual settings.

Employment

Apr 2026 – Present
AI Researcher. Saigon AI Hub, VNG Group.
Serving as *Principal Investigator* for a granted research project on structured reasoning and training strategies for LLMs, funded by VNG Group and Vietnam National University Ho Chi Minh City (VNU-HCM).

Oct 2025 – Present
AI Researcher. Center for AI Research (CAIR), VinUniversity.
Investigating the training of lexical–dense and hybrid retrieval models and their integration into LLM-based QA systems, mentored by Prof. Luu Anh Tuan (NTU, Singapore) and Dr. Nguyen Tran Cong Duy (CAIR).

Dec 2023 – Present
Research Assistant. URA Research Group, Faculty of Computer Science and Engineering, HCMUT.
Conducting research on IR, QA, and LLM-based systems under the supervision of Prof. Quan Thanh Tho, and serving as *Team Leader*, defining research directions and mentoring junior researchers.

Education

2023 – 2027
B.Eng. in Computer Science (High-Quality Program), Ho Chi Minh City University of Technology (HCMUT), VNU-HCM, Vietnam. CGPA: 3.6/4.0.

Selected Publications (full list at https://longstnguyen.com/publications/)

2026

Long S. T. Nguyen, Dat T. Huynh, Quynh T. N. Vo, Dang Le, Cong-Duy T. Nguyen, Anh Tuan Luu, Anh Nguyen-Duc, and Tho T. Quan. *Fast and Focused: Interactive Repository-Level Fill-in-the-Middle Code Completion with Hybrid Structural Retrieval.* Conference on Empirical Methods in Natural Language Processing (**EMNLP**). (CORE A*)

Thi Ty Nguyen, **Long S. T. Nguyen**, Phat T. Tran-Truong, Tan Sang Nguyen, and Tho T. Quan. *Bahnar–Vietnamese Cross-Lingual Sentence Retrieval: A Low-Resource Case Study with Lexical, Neural, and Hybrid Methods.* Asia-Pacific Chapter of the Association for Computational Linguistics & International Joint Conference on Natural Language Processing (**AACL-IJCNLP**).

Long S. T. Nguyen and Tho T. Quan. *Which Works Best for Vietnamese? A Practical Study of Information Retrieval Methods across Domains.* Findings of the Association for Computational Linguistics: **EACL**. (CORE A)

Hung C. Luu, **Long S. T. Nguyen**, Trung M. Pham, Hieu M. Pham, and Tho T. Quan. *HiGraAgent: Dual-Agent Adaptive Reasoning over Hierarchical Knowledge Graph for Open-Domain Multi-hop Question Answering.* Findings of the Association for Computational Linguistics: **EACL**. (CORE A)

Quoc Phu Dang*, Phat T. Tran-Truong*, Duc-Ly Vu, **Long S. T. Nguyen**, Quynh T. N. Vo, and Tho T. Quan. *Enhancing Large Language Model Performance for Automatic Zero-Shot Multiple-Choice Question Answering via Single-Token Logit Prompting.* Computers and Education: Artificial Intelligence. (SCIE, Q1, #1 IN AI)

2025

Long S. T. Nguyen, Quynh T. N. Vo, Hung C. Luu, and Tho T. Quan. *When in Doubt, Ask First: A Unified Retrieval Agent-Based System for Ambiguous and Unanswerable Question Answering.* Findings of the Association for Computational Linguistics: **IJCNLP-AACL**.

Long S. T. Nguyen, Tran T. B. Le, Huong P. N. Nguyen, Quynh T. N. Vo, Phong H. N. Nguyen, and Tho T. Quan. *Serving the Underserved: Leveraging BARTBahnar Language Model for Bahnaric-Vietnamese Translation.* Workshop on Language Models for Underserved Communities (**LM4UC**), at **NAACL**. (WORKSHOP, CORE A)

Academic Service

Prog. Cmte. — AAMAS 2027 (CORE A); AAAI/EAAI 2027 (CORE A*); ACM WSDM 2027 (CORE A).

Conf. Rev. — ACM MM 2026 (CORE A*).

Journal Rev. — IEEE Transactions on Neural Networks and Learning Systems (Q1); Expert Systems with Applications (Q1); Engineering Applications of Artificial Intelligence (Q1); IEEE Access (Q1); Behaviour & Information Technology (Q1); Journal of Natural Language Processing (Q1).

Olympiad — Technical and training support for VOAI 2025 and the preparation of Vietnam's IOAI 2025 team.

References

Supervisor — **Prof. Quan Thanh Tho.** Dean, Faculty of Computer Science and Engineering, HCMUT. qttho@hcmut.edu.vn

Mentor — **Prof. Luu Anh Tuan.** Nanyang Technological University (NTU), Singapore. anhtuan.luu@ntu.edu.sg